



IEEE

DIY Flashlight Workshop

Definitions

What is a voltage?

Voltage

“The measurement of the amount of charge between two points”

- Opposite charges attract;
Like charges repel
- This “push” of the charges trying to repel each other is **voltage!**



Voltage

Where does the voltage
come from?

Voltage Supply

C Battery Voltage Chart		
Battery Type	Full Charge Voltage	Minimum Usable Voltage
Alkaline C	1.5V	1.0V
NiMH C (Rechargeable)	1.2V	1.0V
Lithium C	1.8V	1.2V
Carbon Zinc C	1.5V	0.9V

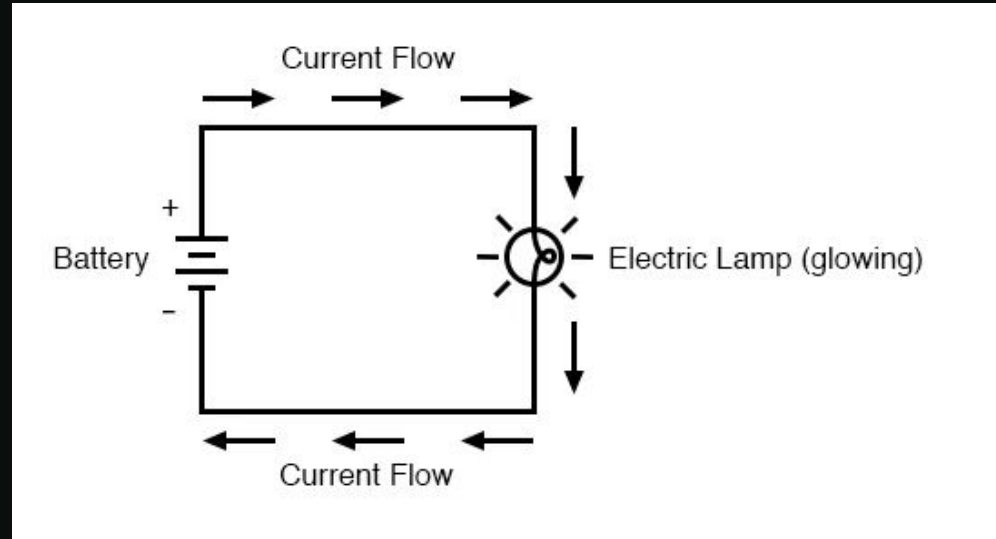
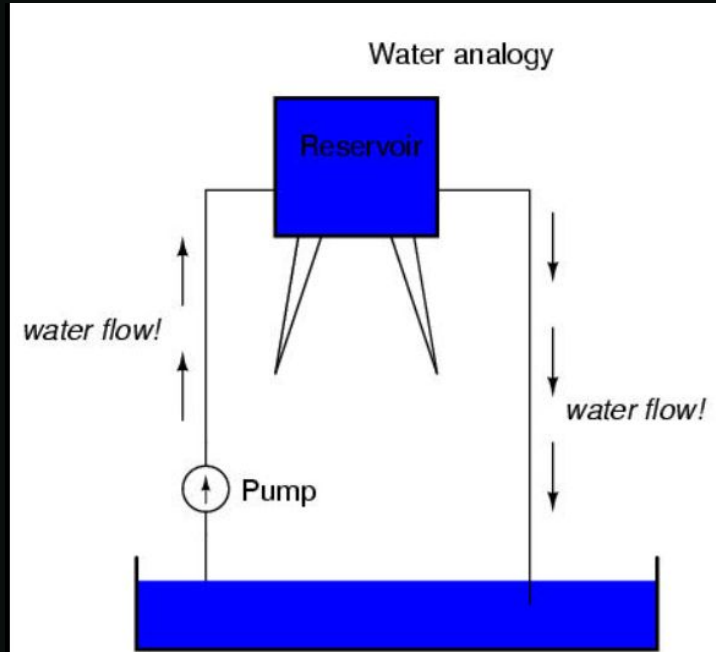
Batteries have a specified voltage used for circuits

One may call it... its
Potential energy

- Terminals
(Negative and Positive Sides)
- Unit of Voltage
Volts (V)
- Voltage loss in
circuit, not created

What is Current?

Case Study: Water Flow

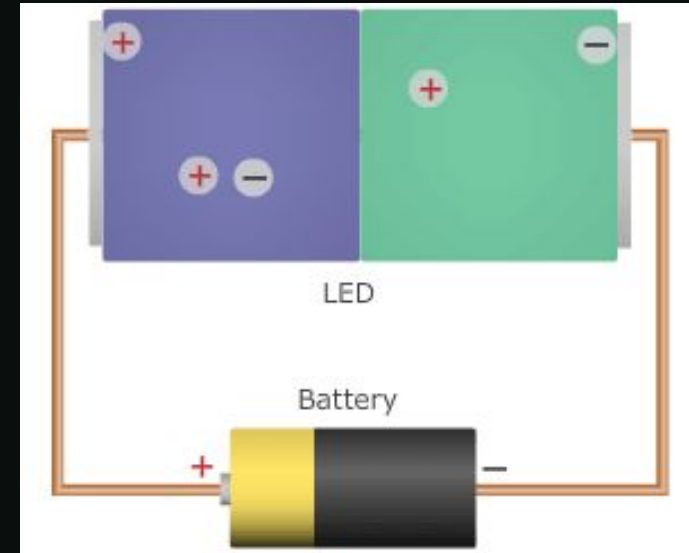
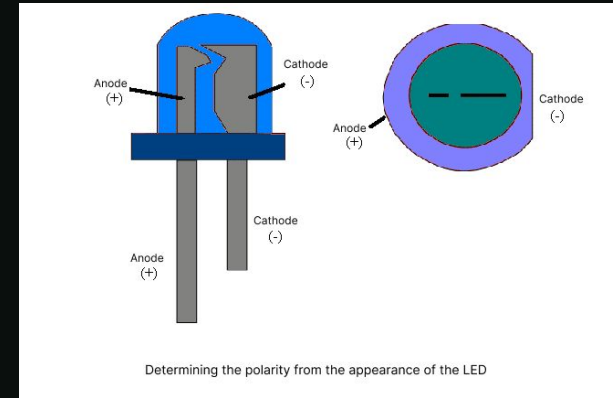


What does water have to do with electrical current?

What does LED stand for?

LED (Light Emitting Diode)

- Produces light with the movement of electrons
- Diode Property: one direction
Ex. one way street
- Polarity: Positive long leg, negative short leg
- Also requires specific voltage (may lead to damage!)



What We Are Building

Battery (Chemical to Electrical) | 3 Volts

LED (Electrical to Light) | 2 Volts with 15 mA

But 3 Volts > 2 Volts

How can we fix this ?

What is Resistance?

Resistance

“Hinders the electric current in a circuit”

- Higher Resistance = Higher Voltage Loss
- Lower Resistance = Lower Voltage Loss



How much resistance do we need?

Ohm's Law

Voltage = Current * Resistance

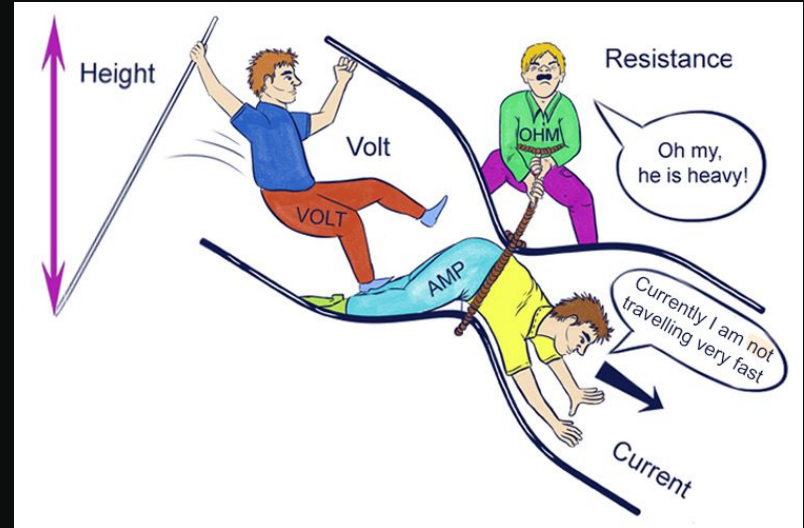
$(1\text{ V}) = (15\text{ mA}) * R$

$15\text{ mA} = 0.015\text{ A}$

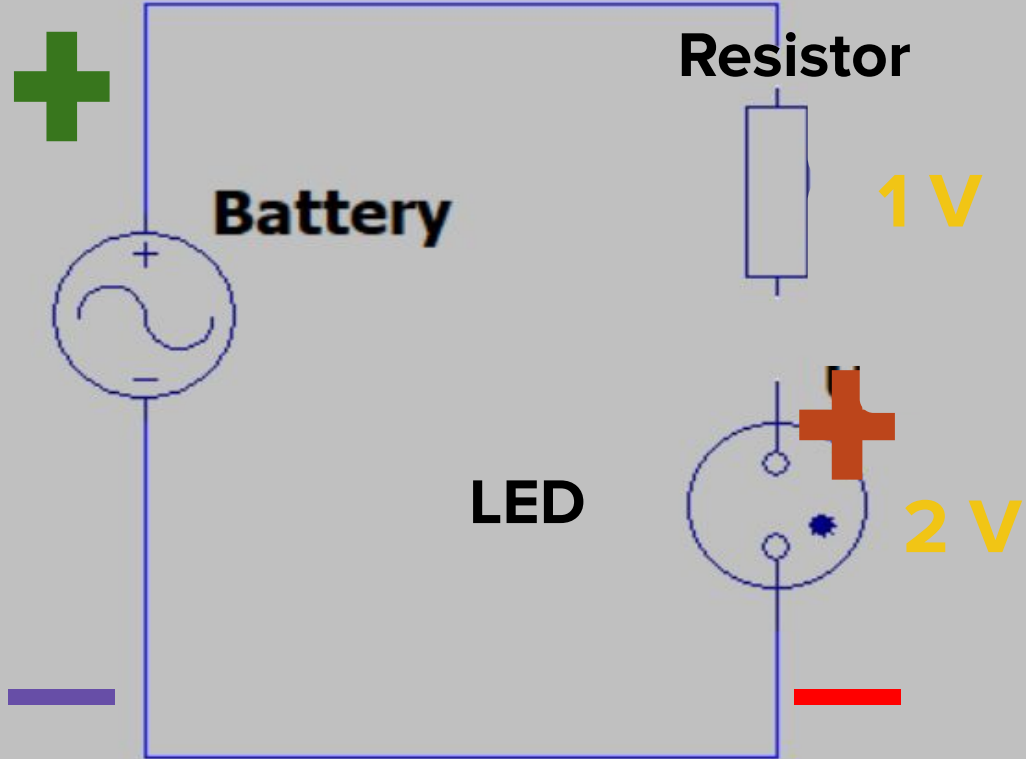
$R = 1\text{ V} / 0.015\text{ A}$

$= \sim 70$
ohms

$R =$ Round up to 100 ohms



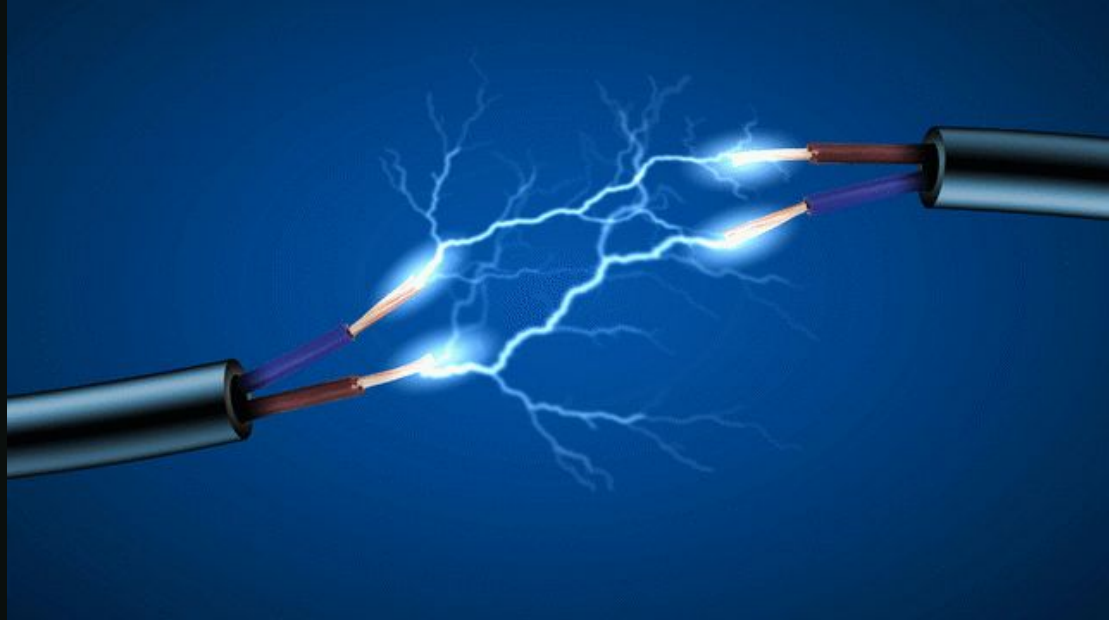
3 V



What are we building?

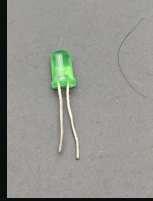
What is the unit for *current*

Amps



Materials

2x Clear 5mm LEDs



2x 5mm LED Holder - Chrome



1x 100 Ohm resistors



1x 3V CR2032 battery and holder

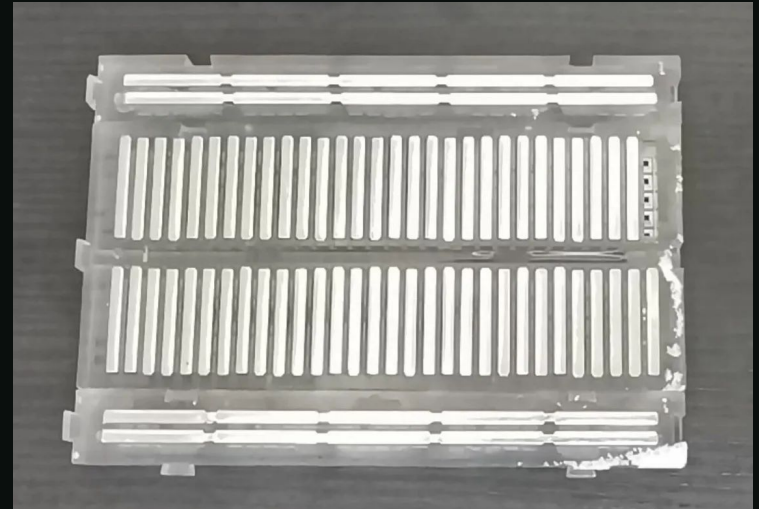
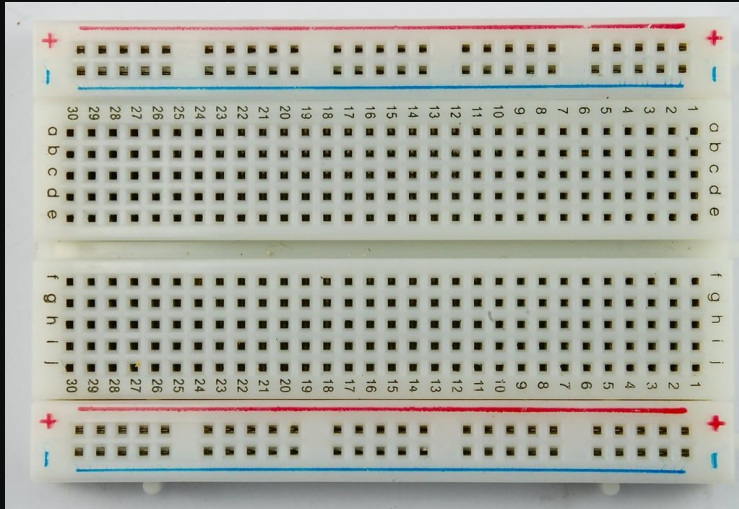


1x Plastic container

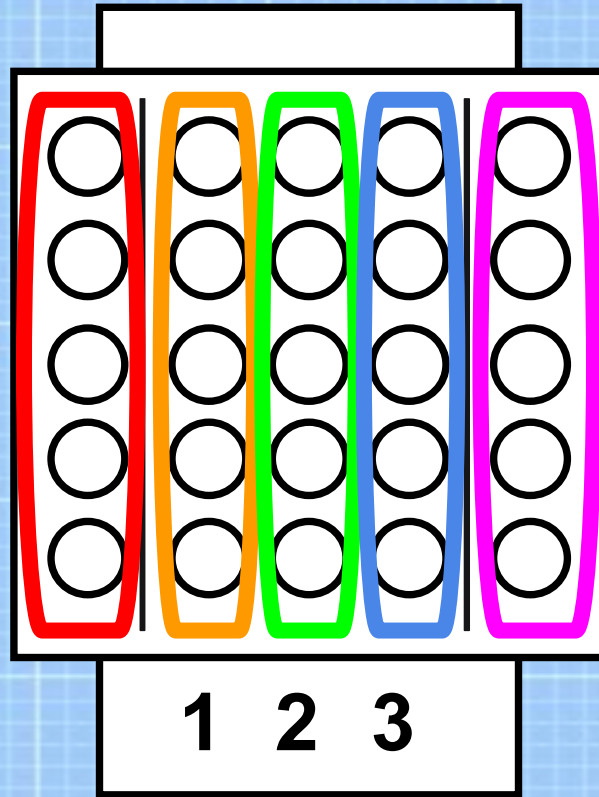


How do breadboard work?

- A typical breadboard (front)
- A typical breadboard (back)



Our Breadboard



Let's Make
Flashlights!



Step 1 – Prepare the container

Screw on LED holders



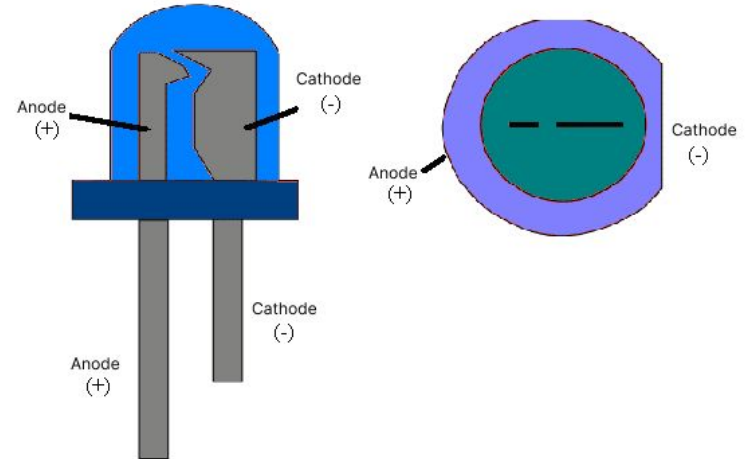
Cathodes & Anodes

Anode (+)

- Positive side
- Long leg

Cathode (-)

- Negative side
- Short leg



Determining the polarity from the appearance of the LED

Step 2 – Prepare the LEDs

1. Insert both LED legs through the plastic holder.
2. Pull shorter leg (-) up by 90 degrees (this leg is the negative side of the LED)

Pay attention to which side is negative!



Step 3 – Connect the LEDs

1. Connect short leg (-) of LED to left column of board.
2. Connect long leg (+) to the right column of board.

(Important: LEDs only work when electricity flows in the right direction.)

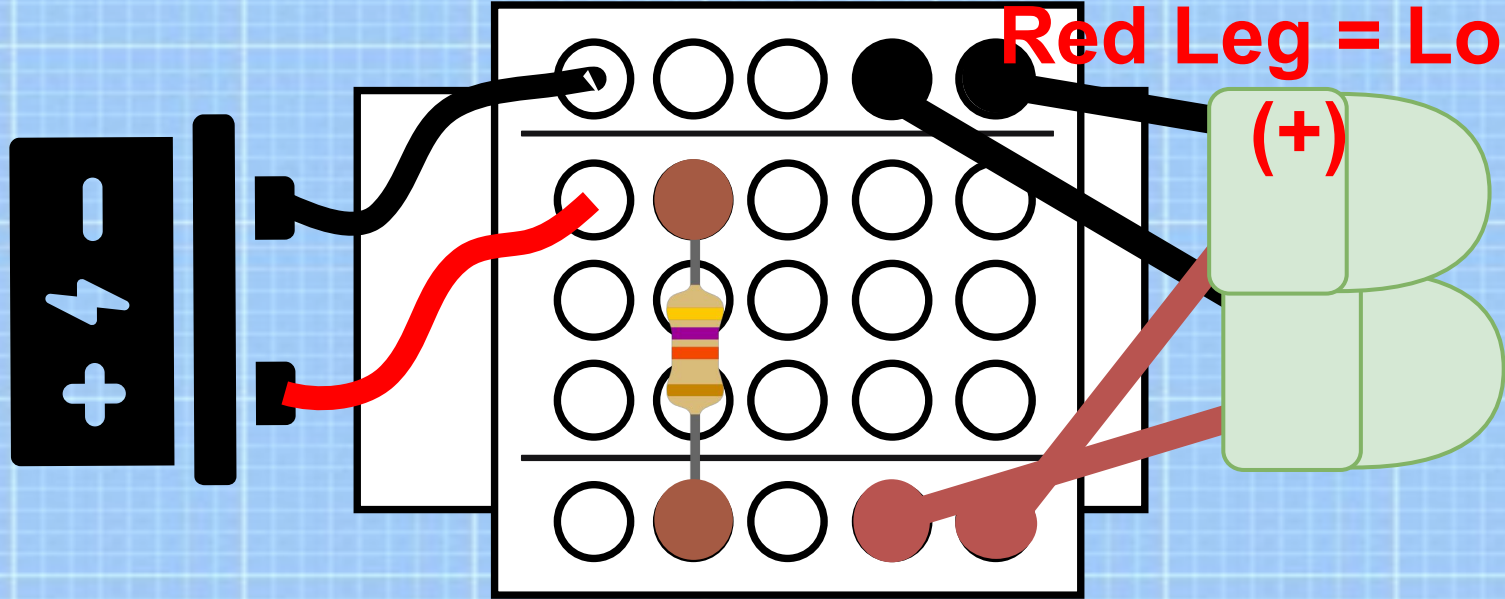
Note: Colors of wire shown might not coordinate with your own.



**Add in the battery.
And the resistor**

**Black Leg =
Short (-)**

**Red Leg = Long
(+)**



Step 4 – Assemble

1. Neatly organize your components in a plastic container.
2. Switch the battery connector



Your flashlight is
now complete!



Review

What did we make today?

Flashlight

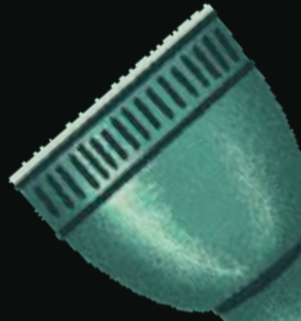
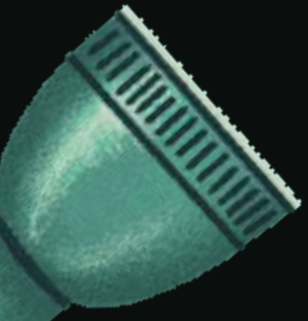
Describe the conversion of energy

Battery: Chemical -> Electrical

LED: Electrical -> Light

The End!

Congratulations on building your very own flashlight!



<https://www.instructables.com/Paper-Circuits/>

